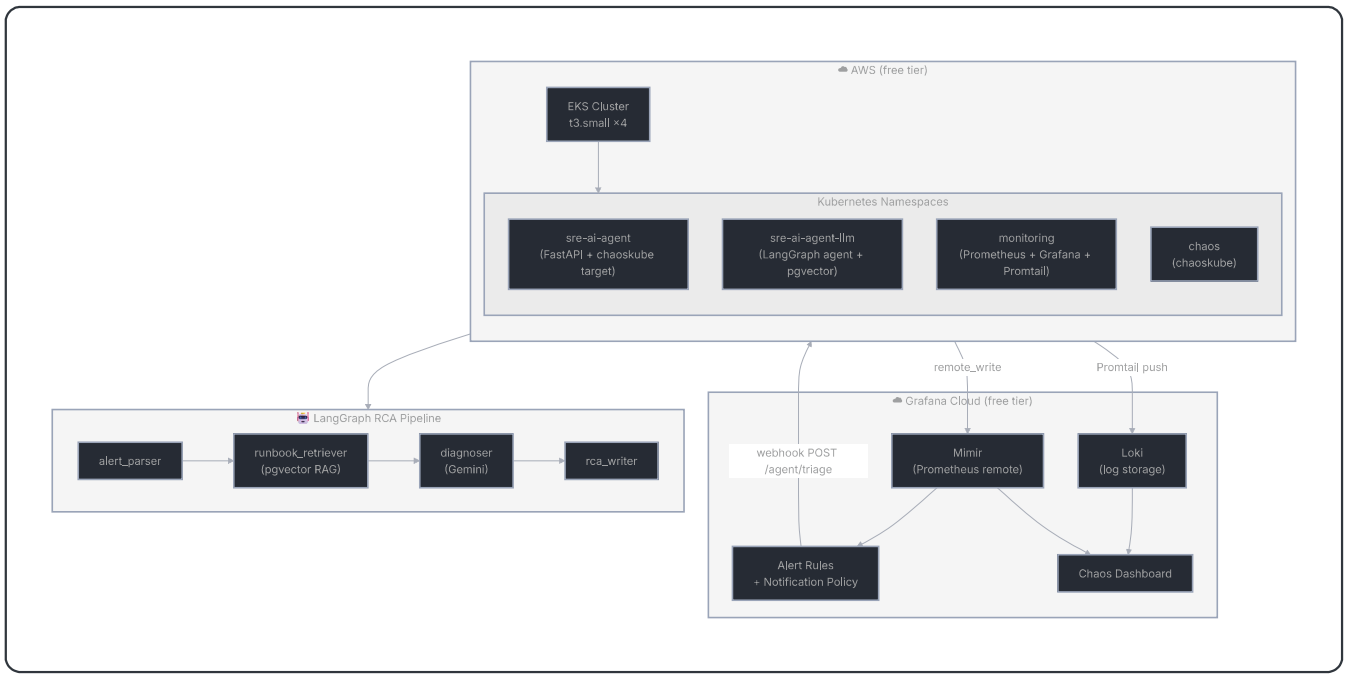
SRE-AI-AGENT-OBSERVABILITY

Complete Deployment Flow

Step-by-step build order — from EKS cluster to live chaos-to-RCA loop

`4 Node Groups · t3.small` `EKS + Grafana Cloud` `LangGraph RCA Agent` `Phase 22 → 24`

SYSTEM OVERVIEW



DEPLOYMENT PHASES

Phase 22-A Infrastructure Bootstrap — `infra-up-complete.sh` ▼

1. EKS Cluster
eksctl creates cluster + VPC + subnets

2. Node Groups
app_nodes (×3 t3.small) + agent_nodes (×1, tainted)

3. IAM Roles
OIDC provider + service account IRSA bindings

4. aws-load-balancer-controller
Helm install — provisions ALBs for services

5. Namespaces
sre-ai-agent, sre-ai-agent-llm, monitoring, chaos

6. Secrets
ECR pull secret, GEMINI_API_KEY, Grafana token

✓ kubectl get nodes → 4 Ready | INSTALL_DATADOG=false | INSTALL_LITMUSCHAOS=false

Phase 22-B pgvector + Runbook Seeding — `deploy-pgvector.sh` ▶

Phase 22-C FastAPI App Deploy — `deploy-app.sh` ▶

Phase 22-D LangGraph Agent Deploy — `deploy-agent.sh` ▶

Phase 22-E Prometheus + Grafana (local) — `infra-up-complete.sh` ▶

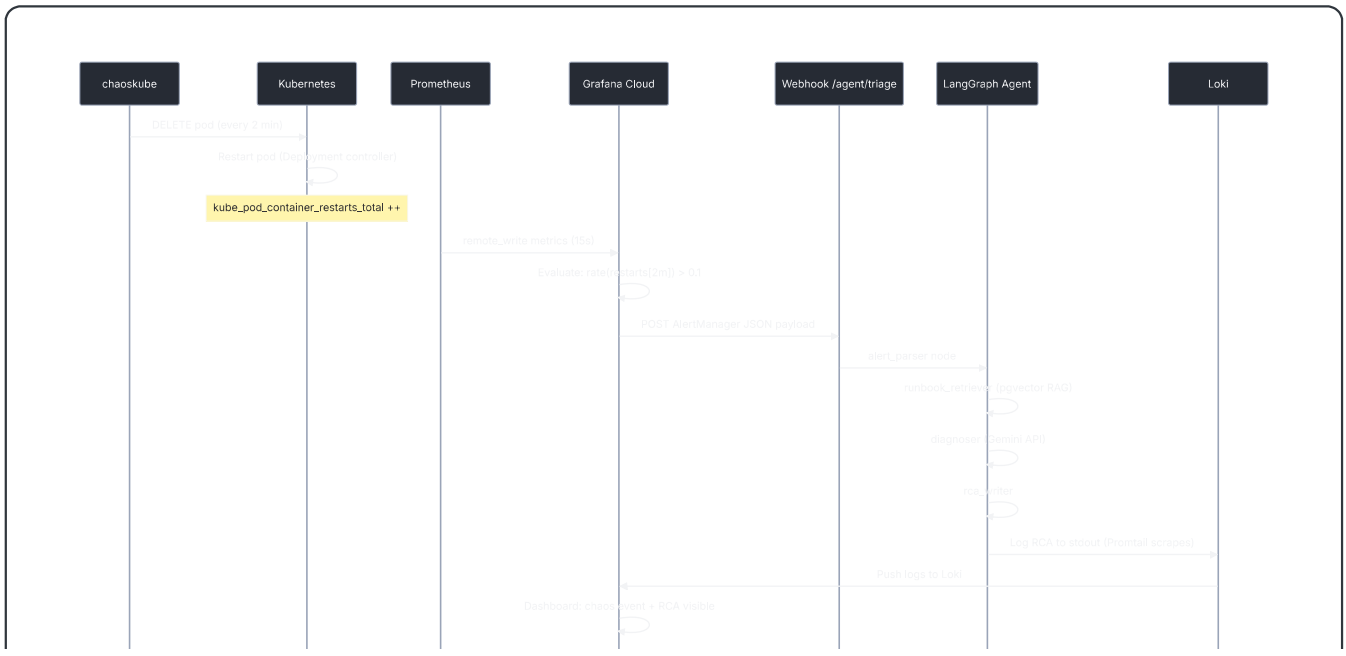
Phase 23 Grafana Cloud + Alerting Webhook — `configure-grafana-cloud.sh` ▶

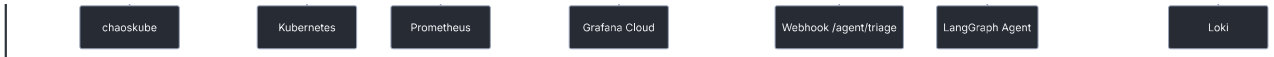
Phase 24-A chaoskube (pod killer) — `deploy-chaoskube.sh` ▶

Phase 24-B Promtail → Grafana Cloud Loki — `deploy-loki-promtail.sh` ▶

Phase 24-C Grafana Cloud Chaos Dashboard — `grafana-chaos-dashboard.json` ▶

LIVE CHAOS-TO-RCA LOOP





KUBERNETES NAMESPACE MAP

sre-ai-agent

- FastAPI app (×3)
- Service (LoadBalancer)
- chaoskube TARGET
- HPA (2–5 replicas)

sre-ai-agent-llm

- LangGraph agent (×1)
- Service (LoadBalancer)
- pgvector StatefulSet
- PVC (10Gi EBS)

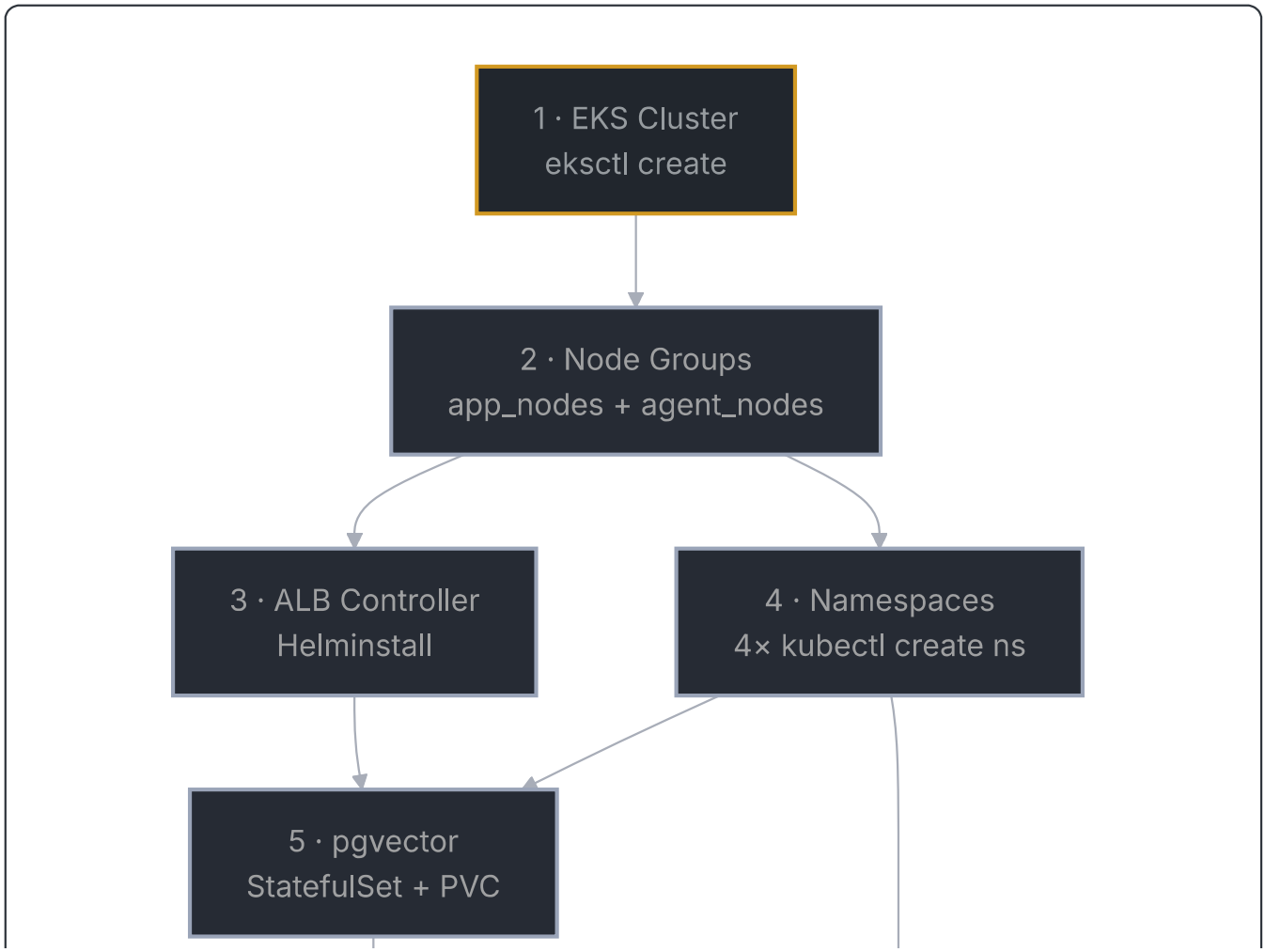
monitoring

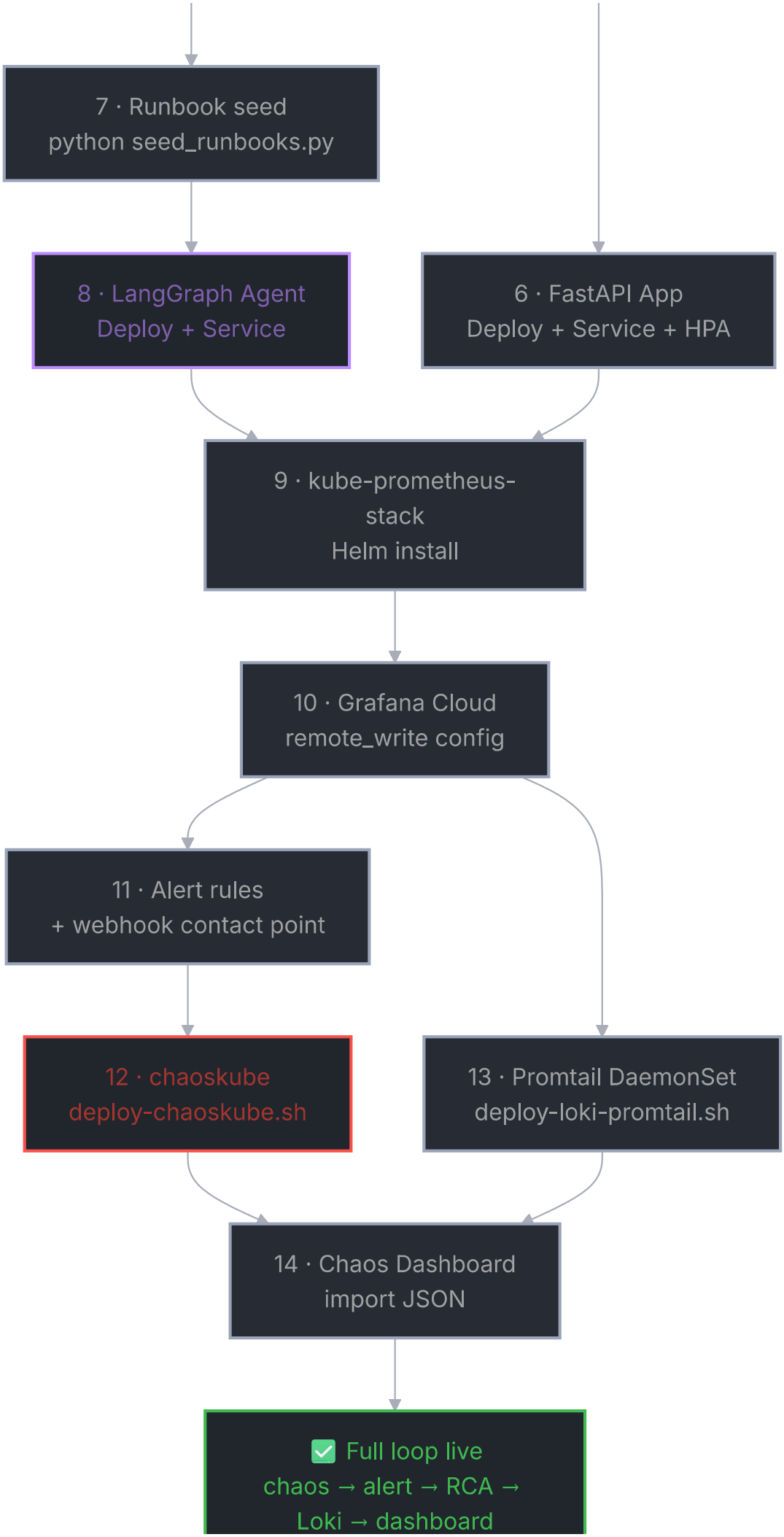
- Prometheus
- Grafana (local)
- AlertManager
- Promtail DaemonSet (×4)

chaos

- chaoskube
- ClusterRole (RBAC)
- ServiceAccount

BUILD ORDER DEPENDENCY GRAPH

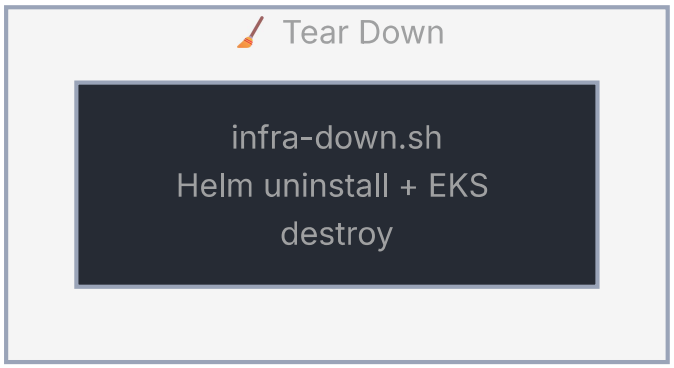


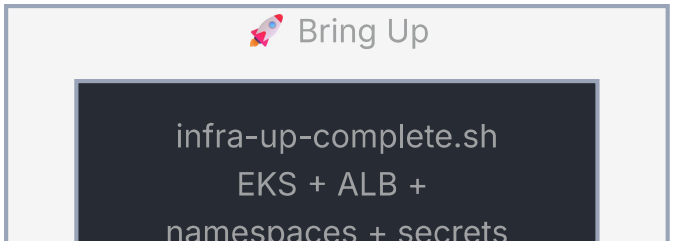

7 · Runbook seed
python seed_runbooks.py
8 · LangGraph Agent
Deploy + Service
6 · FastAPI App
Deploy + Service + HPA
9 · kube-prometheus-stack
Helm install
10 · Grafana Cloud
remote_write config
11 · Alert rules
+ webhook contact point
12 · chaoskube
deploy-chaoskube.sh
13 · Promtail DaemonSet
deploy-loki-promtail.sh
14 · Chaos Dashboard
import JSON
✅ Full loop live
chaos → alert → RCA →
Loki → dashboard

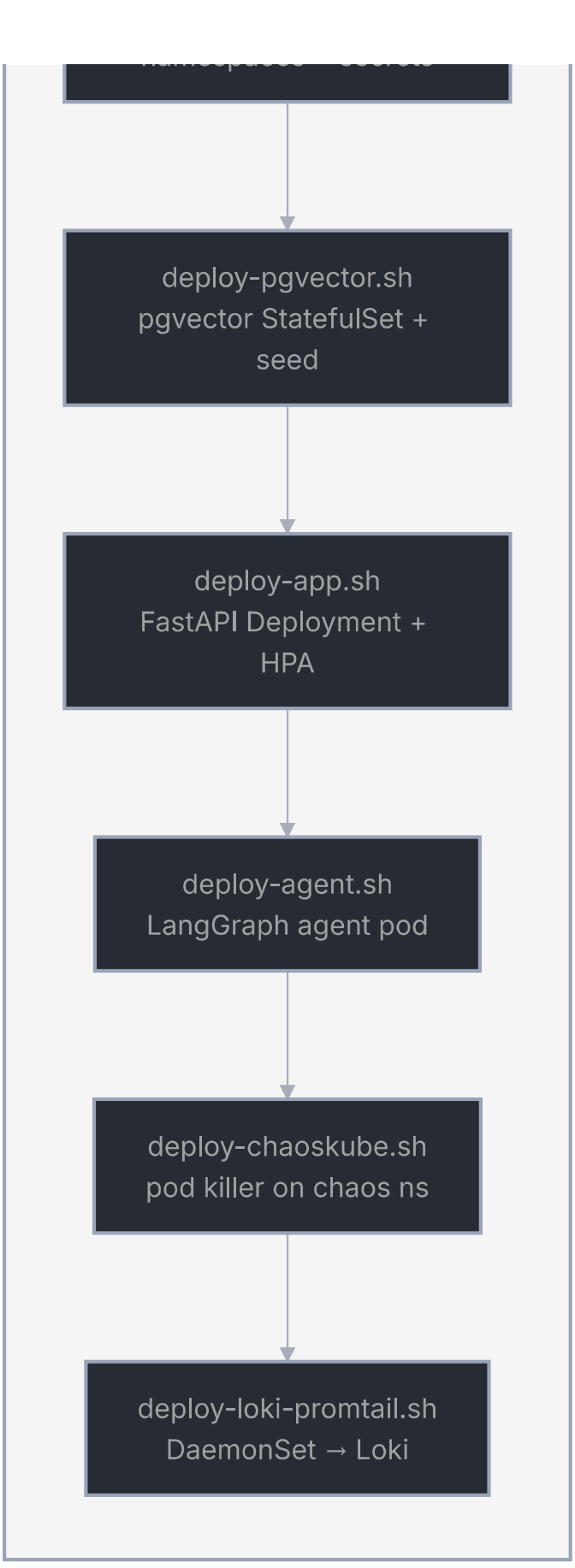
.ENV FILE — ALL REQUIRED KEYS

Key	Example value	Used by
AWS_REGION	ap-south-1	eksctl, ECR, ALB
CLUSTER_NAME	sre-ai-agent-cluster	eksctl, kubectl context
ECR_REGISTRY	123456.dkr.ecr.ap-south-1.amazonaws.com	docker build, Helm values
GEMINI_API_KEY	AIza…	LangGraph diagnoser node
GRAFANA_CLOUD_API_KEY	glc_…	Prometheus remote_write, Promtail
LOKI_USER_ID	1800194	deploy-loki-promtail.sh Basic Auth
GRAFANA_CLOUD_PROMETHEUS_URL	https://prometheus-prod-…/api/prom/push	Prometheus remote_write
INSTALL_DATADOG	false	infra-up-complete.sh flag
INSTALL_LITMUSCHAOS	false	infra-up-complete.sh flag

SCRIPTS AT A GLANCE






deploy-pgvector.sh
pgvector StatefulSet + seed
deploy-app.sh
FastAPI Deployment + HPA
deploy-agent.sh
LangGraph agent pod
deploy-chaoskube.sh
pod killer on chaos ns
deploy-loki-promtail.sh
DaemonSet → Loki